

EMILY DEEB

Spatial Data Scientist | Urban Analytics | MSc Urban Spatial Science, CASA UCL
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PROFESSIONAL SUMMARY

Spatial data scientist and urban analyst with five years' experience, including city-scale geospatial analysis and statistical modelling. At CACI Ltd, I used Python-based machine learning and data analysis for national retail spending projects. Skilled in building data pipelines, stakeholder communication, and translating spatial findings into actionable insights. Seeking opportunities in spatial data science, urban analytics, or location intelligence.

EDUCATION

- 2025 – 2026* **MSc Urban Spatial Science** | The Centre for Advanced Spatial Analysis (CASA), UCL
- Specialising in computational urban analysis, spatial data science, geocomputation, statistical modelling, urban simulation, data visualisation, and machine learning.
- 2015 – 2020* **B.Arch. Architecture** | Bezalel Academy of Arts and Design Jerusalem
- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation. Nominated for David Azrieli Architecture Prize.

PROFESSIONAL EXPERIENCE

- June 2026* **Buro Happold, London** | Industry Consultant — Building a Better Future Scheme
- Led research scoping and team coordination across a multidisciplinary group of six; managed stakeholder expectations, synthesised meeting outputs, and contributed to the final written report and client presentation delivered to Buro Happold and industry stakeholders.
- 2020 – 2025* **Derman Verbakel Architecture, Tel Aviv** | Team Leader — Urban Design & GIS
- Led a team of 4 architects across 15 concurrent projects spanning block to city scale for municipal and private clients including Eilat Municipality and the Israeli Authority for Urban Renewal.
 - Developed and led geo-computational workflows in QGIS and ArcGIS for morphological mapping, site feasibility analysis, market analysis, and statistical profitability modelling; improving office efficiency by 30%.
 - Independently collected, cleaned, and aggregated spatial datasets from scratch for Sakhnin city centre regeneration; delivered evidence-based spatial reports and cartographic visualisations to municipal planning committees.
 - Trained 15+ team members in GIS tools; developed reproducible, documented spatial workflows adopted across the office.
 - Key projects: Transportation Masterplan of Eilat; Regeneration of Sakhnin City Centre (1,200 units); TOD Regeneration Plan Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).
- 2019 – 2020* **SYL Studio Yigal Levi Architecture, Jerusalem** | Architect Intern
- Gained foundational experience in architectural design, documentation, and site coordination.

SELECTED PROJECTS

For full portfolio: <https://emilydeeb.com/>

2026 **Retail Spending & Urban Morphology Prediction** | Industry Dissertation
CACI Ltd x GeoDS Programme

- Investigated whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition, using CACI proprietary consumer spending datasets and HiMoC morphometric classifications.
- Built Python-based machine learning pipeline using GeoPandas, scikit-learn, SHAP, and spatial autocorrelation analysis; applied pre-modelling diagnostics and statistical modelling to produce reproducible, documented analytical outputs.
- Communicated findings through data visualisation and written reporting to CACI's Consumer and Location Intelligence team; presented at GeoDS Data Partner Forum, Saïd Business School, Oxford.

2025 **Food Shock — Global Food Supply Crisis Tracker** | Group Project — React, Leaflet, FAO Data, Python

- Built an interactive web application visualising global food supply chain disruptions using FAO trade flow data (200 countries, 82 commodities), commodity price indices, and spatial mapping.

2025 **London Crime Density & Urban Amenities Analysis** | Python, Machine Learning, Spatial Analysis, London Datastore

- Analysed 1.78 million Metropolitan Police Service crime records across 679 London wards to investigate relationships between urban amenity distribution and crime density.
- Applied multiple regression, K-Nearest Neighbours, and Random Forest models with SHAP decomposition; found nightlife amenities strongest predictors of person/society crime, retail nodes driving property crime.

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy, Matplotlib, GeoPandas, Scikit-learn), R (RStudio), SQL, React.

Data Science & ML: Classification, Clustering, Regression, Statistical Analysis, SHAP, Spatial Autocorrelation, Pre-modelling Diagnostics, Predictive Modelling

Data & Viz: Data Pipelines, Reproducible Workflows, Data Visualisation, Power BI, Quarto, Jupyter, Git

Spatial & GIS: QGIS, ArcGIS, GeoPandas, Spatial Network Analysis, Spatial Data Science, Urban Morphology, Google Earth Engine API

Design & Other: AutoCAD, Rhinoceros, Adobe Suite (Photoshop, Illustrator, InDesign), Microsoft Office, Excel, UI/UX (CalArts certified)

LANGUAGES & CERTIFICATIONS

English (fluent), Hebrew (fluent), Arabic (native)

Licensed Architect — Israeli Labor Department, 2025